

Creation Date 29-Jan-2018

Revision Date 24-Mar-2024

Revision Number 3

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product identifier

Product Description: **Ethanol, anhydrous, denatured**
Cat No. : **C46368**

Unique Formula Identifier (UFI) **6H1F-H4KV-AX0R-P68V**

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use	Laboratory chemicals.
Sector of use	SU3 - Industrial uses: Uses of substances as such or in preparations at industrial sites
Product category	PC21 - Laboratory chemicals
Process categories	PROC15 - Use as a laboratory reagent
Environmental release category	ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates)
Uses advised against	No Information available

1.3. Details of the supplier of the safety data sheet

Company

Thermo Fisher (Kandel) GmbH
Erlenbachweg 2, 76870 Kandel, Germany
Tel: +49 (0) 721 84007 280
Fax: +49 (0) 721 84007 300

Swiss distributor - Fisher Scientific AG
Neuhofstrasse 11, CH 4153 Reinach
Tel: +41 (0) 56 618 41 11
<https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html>

E-mail address **begel.sdsdesk@thermofisher.com**

1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99
CHEMTREC Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:
Tox Info Suisse Emergency Number: **145 (24hr)**
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)
Chemtrec (24h) Toll-Free: 0800 564 402
Chemtrec Local: +41-43 508 20 11 (Zurich)

Poison Centre - Emergency information services

Ireland : National Poisons Information Centre (NPIC) -
01 809 2166 (8am-10pm, 7 days a week)
Malta : +356 2395 2000
Cyprus : +357 2240 5611

SECTION 2: HAZARDS IDENTIFICATION

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2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

Flammable liquids

Category 2 (H225)

Health hazards

Serious Eye Damage/Eye Irritation

Category 2 (H319)

Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

2.2. Label elements



Signal Word

Danger

Hazard Statements

H225 - Highly flammable liquid and vapor

H319 - Causes serious eye irritation

EUH066 - Repeated exposure may cause skin dryness or cracking

Precautionary Statements

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P264 - Wash face, hands and any exposed skin thoroughly after handling

P280 - Wear eye protection/ face protection

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P337 + P313 - If eye irritation persists: Get medical advice/attention

2.3. Other hazards

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB)

Contains a known or suspected endocrine disruptor

Contains a substance on the National Authorities Endocrine Disruptor Lists

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

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3.2. Mixtures

Component	CAS No	EC No	Weight %	CLP Classification - Regulation (EC) No 1272/2008
Ethyl alcohol	64-17-5	200-578-6	>95	Flam. Liq. 2 (H225) Eye Irrit. 2 (H319)
Methyl ethyl ketone	78-93-3	EEC No. 201-159-0	1	Flam. Liq. 2 (H225) Eye Irrit. 2 (H319) STOT SE 3 (H336) (EUH066)
Isopropyl alcohol	67-63-0	200-661-7	1	Flam. Liq. 2 (H225) Eye Irrit. 2 (H319) STOT SE 3 (H336)
Bitrex	3734-33-6	EEC No. 223-095-2	0.001	Acute Tox. 4 (H302) Skin Irrit. 2 (H315) Eye Irrit. 2 (H319) STOT SE 3 (H335)

Component	Specific concentration limits (SCL's)	M-Factor	Component notes
Ethyl alcohol	Eye Irrit. 2 :: C>=50%	-	-

Components	Reach Registration Number
Ethanol	01-2119457610-43
Methyl ethyl ketone	01-2119457290-43
Propan-2-ol	01-2119457558-25
Denatonium benzoate	01-2120102843-65

Full text of Hazard Statements: see section 16

SECTION 4: FIRST AID MEASURES

4.1. Description of first aid measures

General Advice	If symptoms persist, call a physician.
Eye Contact	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.
Skin Contact	Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.
Ingestion	Clean mouth with water and drink afterwards plenty of water.
Inhalation	Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.
Self-Protection of the First Aider	Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

4.2. Most important symptoms and effects, both acute and delayed

Difficulty in breathing. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician	Treat symptomatically. Symptoms may be delayed.
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SECTION 5: FIREFIGHTING MEASURES

5.1. Extinguishing media

Suitable Extinguishing Media

Water spray, carbon dioxide (CO₂), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

Extinguishing media which must not be used for safety reasons

Do not use a solid water stream as it may scatter and spread fire.

5.2. Special hazards arising from the substance or mixture

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

Hazardous Combustion Products

None under normal use conditions.

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges.

6.2. Environmental precautions

Do not flush into surface water or sanitary sewer system.

6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before

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re-use. Wash hands before breaks and after work.

7.2. Conditions for safe storage, including any incompatibilities

Keep away from heat, sparks and flame. Flammables area. Keep container tightly closed in a dry and well-ventilated place.

Technical Rules for Hazardous Substances (TRGS) 510
Storage Class (LGK) (Germany)

Class 3

Switzerland - Storage of hazardous substances

Storage class - SC 3

<https://www.kvu.ch/de/themen/stoffe-und-produkte>

<https://www.kvu.ch/fr/themes/substances-et-produits>

<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

7.3. Specific end use(s)

Use in laboratories

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Exposure limits

List source(s): **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

Component	European Union	The United Kingdom	France	Belgium	Spain
Ethyl alcohol		TWA: 1000 ppm TWA; 1920 mg/m ³ TWA WEL - STEL: 3000 ppm STEL: 5760 mg/m ³ STEL	TWA / VME: 1000 ppm (8 heures). TWA / VME: 1900 mg/m ³ (8 heures). STEL / VLCT: 5000 ppm. STEL / VLCT: 9500 mg/m ³ .	TWA: 1000 ppm 8 uren TWA: 1907 mg/m ³ 8 uren	STEL / VLA-EC: 1000 ppm (15 minutos). STEL / VLA-EC: 1910 mg/m ³ (15 minutos).
Methyl ethyl ketone	TWA: 200 ppm (8h) TWA: 600 mg/m ³ (8h) STEL: 300 ppm (15min) STEL: 900 mg/m ³ (15min)	STEL: 300 ppm 15 min STEL: 899 mg/m ³ 15 min TWA: 200 ppm 8 hr TWA: 600 mg/m ³ 8 hr Skin	TWA / VME: 200 ppm (8 heures). restrictive limit TWA / VME: 600 mg/m ³ (8 heures). restrictive limit STEL / VLCT: 300 ppm. restrictive limit STEL / VLCT: 900 mg/m ³ . restrictive limit Peau	TWA: 200 ppm 8 uren TWA: 600 mg/m ³ 8 uren STEL: 300 ppm 15 minuten STEL: 900 mg/m ³ 15 minuten	STEL / VLA-EC: 300 ppm (15 minutos). STEL / VLA-EC: 900 mg/m ³ (15 minutos). TWA / VLA-ED: 200 ppm (8 horas) TWA / VLA-ED: 600 mg/m ³ (8 horas)
Isopropyl alcohol		STEL: 500 ppm 15 min STEL: 1250 mg/m ³ 15 min TWA: 400 ppm 8 hr TWA: 999 mg/m ³ 8 hr	STEL / VLCT: 400 ppm. STEL / VLCT: 980 mg/m ³ .	TWA: 200 ppm 8 uren TWA: 500 mg/m ³ 8 uren STEL: 400 ppm 15 minuten STEL: 1000 mg/m ³ 15 minuten	STEL / VLA-EC: 400 ppm (15 minutos). STEL / VLA-EC: 1000 mg/m ³ (15 minutos). TWA / VLA-ED: 200 ppm (8 horas) TWA / VLA-ED: 500 mg/m ³ (8 horas)

Component	Italy	Germany	Portugal	The Netherlands	Finland
Ethyl alcohol		200 ppm TWA MAK; 380 mg/m ³ TWA MAK	STEL: 1000 ppm 15 minutos	huid STEL: 1900 mg/m ³ 15 minuten TWA: 260 mg/m ³ 8 uren	TWA: 1000 ppm 8 tunteina TWA: 1900 mg/m ³ 8 tunteina

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					STEL: 1300 ppm 15 minuutteina STEL: 2500 mg/m ³ 15 minuutteina
Methyl ethyl ketone	TWA: 200 ppm 8 ore. Time Weighted Average TWA: 600 mg/m ³ 8 ore. Time Weighted Average STEL: 300 ppm 15 minuti. Short-term STEL: 900 mg/m ³ 15 minuti. Short-term	TWA: 200 ppm (8 Stunden). AGW - exposure factor 1 TWA: 600 mg/m ³ (8 Stunden). AGW - exposure factor 1 TWA: 200 ppm (8 Stunden). MAK TWA: 600 mg/m ³ (8 Stunden). MAK Höhepunkt: 200 ppm Höhepunkt: 600 mg/m ³ Haut	STEL: 300 ppm 15 minutos STEL: 900 mg/m ³ 15 minutos TWA: 200 ppm 8 horas TWA: 600 mg/m ³ 8 horas	huid STEL: 900 mg/m ³ 15 minuten TWA: 590 mg/m ³ 8 uren	TWA: 20 ppm 8 tunteina TWA: 60 mg/m ³ 8 tunteina STEL: 100 ppm 15 minuutteina STEL: 300 mg/m ³ 15 minuutteina Iho
Isopropyl alcohol		TWA: 200 ppm (8 Stunden). AGW - exposure factor 2 TWA: 500 mg/m ³ (8 Stunden). AGW - exposure factor 2 TWA: 200 ppm (8 Stunden). MAK TWA: 500 mg/m ³ (8 Stunden). MAK Höhepunkt: 400 ppm Höhepunkt: 1000 mg/m ³	STEL: 400 ppm 15 minutos TWA: 200 ppm 8 horas		TWA: 200 ppm 8 tunteina TWA: 500 mg/m ³ 8 tunteina STEL: 250 ppm 15 minuutteina STEL: 620 mg/m ³ 15 minuutteina

Component	Austria	Denmark	Switzerland	Poland	Norway
Ethyl alcohol	MAK-KZGW: 2000 ppm 15 Minuten MAK-KZGW: 3800 mg/m ³ 15 Minuten MAK-TMW: 1000 ppm 8 Stunden MAK-TMW: 1900 mg/m ³ 8 Stunden	TWA: 1000 ppm 8 timer TWA: 1900 mg/m ³ 8 timer STEL: 2000 ppm 15 minutter STEL: 3800 mg/m ³ 15 minutter	STEL: 1000 ppm 15 Minuten STEL: 1920 mg/m ³ 15 Minuten TWA: 500 ppm 8 Stunden TWA: 960 mg/m ³ 8 Stunden	TWA: 1900 mg/m ³ 8 godzinach	TWA: 500 ppm 8 timer TWA: 950 mg/m ³ 8 timer STEL: 625 ppm 15 minutter. value calculated STEL: 1187.5 mg/m ³ 15 minutter. value calculated
Methyl ethyl ketone	Haut MAK-KZGW: 200 ppm 15 Minuten MAK-KZGW: 590 mg/m ³ 15 Minuten MAK-TMW: 100 ppm 8 Stunden MAK-TMW: 295 mg/m ³ 8 Stunden	TWA: 50 ppm 8 timer TWA: 145 mg/m ³ 8 timer STEL: 900 mg/m ³ 15 minutter STEL: 300 ppm 15 minutter Hud	Haut/Peau STEL: 200 ppm 15 Minuten STEL: 590 mg/m ³ 15 Minuten TWA: 200 ppm 8 Stunden TWA: 590 mg/m ³ 8 Stunden	STEL: 900 mg/m ³ 15 minutach TWA: 450 mg/m ³ 8 godzinach	TWA: 75 ppm 8 timer TWA: 220 mg/m ³ 8 timer STEL: 112.5 ppm 15 minutter. value calculated STEL: 275 mg/m ³ 15 minutter. value calculated
Isopropyl alcohol	MAK-KZGW: 800 ppm 15 Minuten MAK-KZGW: 2000 mg/m ³ 15 Minuten MAK-TMW: 200 ppm 8 Stunden MAK-TMW: 500 mg/m ³ 8 Stunden	TWA: 200 ppm 8 timer TWA: 490 mg/m ³ 8 timer STEL: 400 ppm 15 minutter STEL: 980 mg/m ³ 15 minutter	STEL: 400 ppm 15 Minuten STEL: 1000 mg/m ³ 15 Minuten TWA: 200 ppm 8 Stunden TWA: 500 mg/m ³ 8 Stunden	STEL: 1200 mg/m ³ 15 minutach TWA: 900 mg/m ³ 8 godzinach	TWA: 100 ppm 8 timer TWA: 245 mg/m ³ 8 timer STEL: 150 ppm 15 minutter. value calculated STEL: 306.25 mg/m ³ 15 minutter. value calculated

Component	Bulgaria	Croatia	Ireland	Cyprus	Czech Republic
Ethyl alcohol	TWA: 1000 mg/m ³	TWA-GVI: 1000 ppm 8 satima. TWA-GVI: 1900 mg/m ³ 8 satima.	STEL: 1000 ppm 15 min		TWA: 1000 mg/m ³ 8 hodinách. Ceiling: 3000 mg/m ³
Methyl ethyl ketone	TWA: 590 mg/m ³ STEL : 885 mg/m ³	TWA-GVI: 200 ppm 8 satima. TWA-GVI: 600 mg/m ³ 8 satima. STEL-KGVI: 300 ppm 15 minutama. STEL-KGVI: 900 mg/m ³	TWA: 200 ppm 8 hr. TWA: 600 mg/m ³ 8 hr. STEL: 300 ppm 15 min STEL: 900 mg/m ³ 15 min Skin	STEL: 300 ppm STEL: 900 mg/m ³ TWA: 200 ppm TWA: 600 mg/m ³	TWA: 600 mg/m ³ 8 hodinách. Ceiling: 900 mg/m ³

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		15 minutama.			
Isopropyl alcohol	TWA: 980.0 mg/m ³ STEL : 1225.0 mg/m ³	TWA-GVI: 400 ppm 8 satima. TWA-GVI: 999 mg/m ³ 8 satima. STEL-KGVI: 500 ppm 15 minutama. STEL-KGVI: 1250 mg/m ³ 15 minutama.	TWA: 200 ppm 8 hr. STEL: 400 ppm 15 min Skin		TWA: 500 mg/m ³ 8 hodinách. Potential for cutaneous absorption Ceiling: 1000 mg/m ³

Component	Estonia	Gibraltar	Greece	Hungary	Iceland
Ethyl alcohol	TWA: 500 ppm 8 tundides. TWA: 1000 mg/m ³ 8 tundides. STEL: 1000 ppm 15 minutites. STEL: 1900 mg/m ³ 15 minutites.		TWA: 1000 ppm TWA: 1900 mg/m ³	STEL: 3800 mg/m ³ 15 percekben. CK TWA: 1900 mg/m ³ 8 órában. AK	TWA: 1000 ppm 8 klukkustundum. TWA: 1900 mg/m ³ 8 klukkustundum. Ceiling: 2000 ppm Ceiling: 3800 mg/m ³
Methyl ethyl ketone	TWA: 200 ppm 8 tundides. TWA: 600 mg/m ³ 8 tundides. STEL: 300 ppm 15 minutites. STEL: 900 mg/m ³ 15 minutites.	TWA: 200 ppm 8 hr TWA: 600 mg/m ³ 8 hr STEL: 300 ppm 15 min STEL: 900 mg/m ³ 15 min	STEL: 300 ppm STEL: 900 mg/m ³ TWA: 200 ppm TWA: 600 mg/m ³	STEL: 900 mg/m ³ 15 percekben. CK TWA: 600 mg/m ³ 8 órában. AK lehetséges bőrön keresztüli felszívódás	STEL: 300 ppm STEL: 900 mg/m ³ TWA: 50 ppm 8 klukkustundum. TWA: 145 mg/m ³ 8 klukkustundum. Skin notation
Isopropyl alcohol	TWA: 150 ppm 8 tundides. TWA: 350 mg/m ³ 8 tundides. STEL: 250 ppm 15 minutites. STEL: 600 mg/m ³ 15 minutites.		STEL: 500 ppm STEL: 1225 mg/m ³ TWA: 400 ppm TWA: 980 mg/m ³	STEL: 1000 mg/m ³ 15 percekben. CK TWA: 500 mg/m ³ 8 órában. AK lehetséges bőrön keresztüli felszívódás	TWA: 200 ppm 8 klukkustundum. TWA: 490 mg/m ³ 8 klukkustundum. Skin notation Ceiling: 400 ppm Ceiling: 980 mg/m ³

Component	Latvia	Lithuania	Luxembourg	Malta	Romania
Ethyl alcohol	TWA: 1000 mg/m ³	TWA: 500 ppm IPRD TWA: 1000 mg/m ³ IPRD STEL: 1000 ppm STEL: 1900 mg/m ³			TWA: 1000 ppm 8 ore TWA: 1900 mg/m ³ 8 ore STEL: 5000 ppm 15 minute STEL: 9500 mg/m ³ 15 minute
Methyl ethyl ketone	STEL: 300 ppm STEL: 900 mg/m ³ TWA: 67 ppm TWA: 200 mg/m ³		TWA: 200 ppm 8 Stunden TWA: 600 mg/m ³ 8 Stunden STEL: 300 ppm 15 Minuten STEL: 900 mg/m ³ 15 Minuten	TWA: 200 ppm TWA: 600 mg/m ³ STEL: 300 ppm 15 minuti STEL: 900 mg/m ³ 15 minuti	TWA: 200 ppm 8 ore TWA: 600 mg/m ³ 8 ore STEL: 300 ppm 15 minute STEL: 900 mg/m ³ 15 minute
Isopropyl alcohol	STEL: 600 mg/m ³ TWA: 350 mg/m ³	TWA: 150 ppm IPRD TWA: 350 mg/m ³ IPRD STEL: 250 ppm STEL: 600 mg/m ³			TWA: 81 ppm 8 ore TWA: 200 mg/m ³ 8 ore STEL: 203 ppm 15 minute STEL: 500 mg/m ³ 15 minute

Component	Russia	Slovak Republic	Slovenia	Sweden	Turkey
Ethyl alcohol	TWA: 1000 mg/m ³ 2391 MAC: 2000 mg/m ³	Ceiling: 1920 mg/m ³ TWA: 500 ppm TWA: 960 mg/m ³	TWA: 960 mg/m ³ 8 urah TWA: 500 ppm 8 urah STEL: 1000 ppm 15 minutah STEL: 1920 mg/m ³ 15 minutah	Indicative STEL: 1000 ppm 15 minuter Indicative STEL: 1900 mg/m ³ 15 minuter TLV: 500 ppm 8 timmar. NGV TLV: 1000 mg/m ³ 8 timmar. NGV	
Methyl ethyl ketone	TWA: 200 mg/m ³ 0421 MAC: 400 mg/m ³	Ceiling: 900 mg/m ³ TWA: 200 ppm	TWA: 200 ppm 8 urah TWA: 600 mg/m ³ 8 urah	Binding STEL: 300 ppm 15 minuter	TWA: 200 ppm 8 saat TWA: 600 mg/m ³ 8 saat

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		TWA: 600 mg/m ³	Koža STEL: 300 ppm 15 minutah STEL: 900 mg/m ³ 15 minutah	Binding STEL: 900 mg/m ³ 15 minuter TLV: 50 ppm 8 timmar. NGV TLV: 150 mg/m ³ 8 timmar. NGV	STEL: 300 ppm 15 dakika STEL: 900 mg/m ³ 15 dakika
Isopropyl alcohol	TWA: 10 mg/m ³ 1761 MAC: 50 mg/m ³	Ceiling: 1000 mg/m ³ TWA: 200 ppm TWA: 500 mg/m ³	TWA: 200 ppm 8 urah TWA: 500 mg/m ³ 8 urah STEL: 400 ppm 15 minutah STEL: 1000 mg/m ³ 15 minutah	Indicative STEL: 250 ppm 15 minuter Indicative STEL: 600 mg/m ³ 15 minuter TLV: 150 ppm 8 timmar. NGV TLV: 350 mg/m ³ 8 timmar. NGV	

Biological limit values

List source(s): **UK** - Biological Monitoring Guidance Values provided by the UK's Health and Safety Executive (HSE) Control of Substances Hazardous to Health Regulations (COSHH) 2002 (as amended) and EH40/2005.

Component	European Union	United Kingdom	France	Spain	Germany
Methyl ethyl ketone		Butan-2-one: 70 µmol/L urine post shift	Methylethylketone: 2 mg/L urine end of shift	Methyl ethyl ketone: 2 mg/L urine end of shift	2-Butanone: 2 mg/L urine (end of shift)
Isopropyl alcohol				Acetone: 40 mg/L urine end of workweek	Acetone: 25 mg/L whole blood (end of shift) Acetone: 25 mg/L urine (end of shift)

Component	Italy	Finland	Denmark	Bulgaria	Romania
Methyl ethyl ketone					Methylethylketone: 2 mg/L urine end of shift
Isopropyl alcohol					Acetone: 50 mg/L urine end of shift

Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS70 General methods for sampling airborne gases and vapours

MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography

MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

Component	Acute effects local (Oral)	Acute effects systemic (Oral)	Chronic effects local (Oral)	Chronic effects systemic (Oral)
Ethyl alcohol 64-17-5 (>95)		DNEL = 87 mg/kg bw/d		

Component	Acute effects local (Dermal)	Acute effects systemic (Dermal)	Chronic effects local (Dermal)	Chronic effects systemic (Dermal)
Ethyl alcohol 64-17-5 (>95)				DNEL = 343mg/kg bw/day
Methyl ethyl ketone 78-93-3 (1)				DNEL = 1161mg/kg bw/day
Isopropyl alcohol 67-63-0 (1)				DNEL = 888mg/kg bw/day
Bitrex 3734-33-6 (0.001)				DNEL = 1.43mg/kg bw/day

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Component	Acute effects local (Inhalation)	Acute effects systemic (Inhalation)	Chronic effects local (Inhalation)	Chronic effects systemic (Inhalation)
Ethyl alcohol 64-17-5 (>95)	DNEL = 1900mg/m ³			DNEL = 950mg/m ³
Methyl ethyl ketone 78-93-3 (1)				DNEL = 600mg/m ³
Isopropyl alcohol 67-63-0 (1)				DNEL = 500mg/m ³
Bitrex 3734-33-6 (0.001)				DNEL = 4.99mg/m ³

Predicted No Effect Concentration (PNEC)

See values below.

Component	Fresh water	Fresh water sediment	Water Intermittent	Microorganisms in sewage treatment	Soil (Agriculture)
Methyl ethyl ketone 78-93-3 (1)	PNEC = 55.8mg/L	PNEC = 284.74mg/kg sediment dw	PNEC = 55.8mg/L	PNEC = 709mg/L	PNEC = 22.5mg/kg soil dw
Isopropyl alcohol 67-63-0 (1)	PNEC = 140.9mg/L	PNEC = 552mg/kg sediment dw	PNEC = 140.9mg/L	PNEC = 2251mg/L	PNEC = 28mg/kg soil dw
Bitrex 3734-33-6 (0.001)	PNEC = 0.1mg/L	PNEC = 25mg/kg sediment dw	PNEC = 1mg/L		PNEC = 4.95mg/kg soil dw

Component	Marine water	Marine water sediment	Marine water Intermittent	Food chain	Air
Methyl ethyl ketone 78-93-3 (1)	PNEC = 55.8mg/L	PNEC = 284.7mg/kg sediment dw		PNEC = 1000mg/kg food	
Isopropyl alcohol 67-63-0 (1)	PNEC = 140.9mg/L	PNEC = 552mg/kg sediment dw		PNEC = 160mg/kg food	
Bitrex 3734-33-6 (0.001)	PNEC = 10µg/L	PNEC = 2.5mg/kg sediment dw	PNEC = 0.1mg/L		

8.2. Exposure controls

Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting/equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Personal protective equipment

Eye Protection

Goggles (European standard - EN 166)

Hand Protection

Protective gloves

Glove material	Breakthrough time	Glove thickness	EU standard	Glove comments
Nitrile rubber	See manufacturers	-	EN 374	(minimum requirement)
Viton (R)	recommendations			

Skin and body protection

Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

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Respiratory Protection	When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly
Large scale/emergency use	Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced Recommended Filter type: Organic gases and vapours filter Type A Brown conforming to EN14387
Small scale/Laboratory use	Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. Recommended half mask:- Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141 When RPE is used a face piece Fit Test should be conducted
Environmental exposure controls	Prevent product from entering drains. Do not allow material to contaminate ground water system.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Physical State	Liquid	
Appearance		
Odor	No information available	
Odor Threshold	No data available	
Melting Point/Range	-114.1 °C / -173.4 °F	
Softening Point	No data available	
Boiling Point/Range	78 °C / 172.4 °F	@ 760 mmHg
Flammability (liquid)	Highly flammable	On basis of test data
Flammability (solid,gas)	Not applicable	Liquid
Explosion Limits	Lower 2.5 Upper 13.5	
Flash Point	12 °C / 53.6 °F	Method - No information available
Autoignition Temperature	363 °C / 685.4 °F	
Decomposition Temperature	No data available	
pH	5.7	
Viscosity	No data available	
Water Solubility	Soluble	
Solubility in other solvents	No information available	
Partition Coefficient (n-octanol/water)		
Component	log Pow	
Ethyl alcohol	-0.32	
Methyl ethyl ketone	0.29	
Isopropyl alcohol	0.05	
Bitrex	2.2	
Vapor Pressure	57 hPa @ 20°C	
Density / Specific Gravity	0.79	
Bulk Density	Not applicable	Liquid
Vapor Density	No data available	(Air = 1.0)
Particle characteristics	Not applicable (liquid)	

9.2. Other information

Explosive Properties	Vapors may form explosive mixtures with air
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SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity

None known, based on information available

10.2. Chemical stability

Stable under normal conditions.

10.3. Possibility of hazardous reactions

Hazardous Polymerization Hazardous Reactions

No information available.
None under normal processing.

10.4. Conditions to avoid

Keep away from open flames, hot surfaces and sources of ignition.

10.5. Incompatible materials

None known.

10.6. Hazardous decomposition products

None under normal use conditions.

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information

(a) acute toxicity;

Oral

Based on available data, the classification criteria are not met

Dermal

Based on available data, the classification criteria are not met

Inhalation

Based on available data, the classification criteria are not met

Toxicology data for the components

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Ethyl alcohol	LD50 = 10470 mg/kg OECD 401 (Rat) 3450 mg/kg (Mouse)	-	LC50 = 117-125 mg/l (4h) OECD 403 (rat) 20000 ppm/10H (rat)
Methyl ethyl ketone	LD50 = 2483 mg/kg (Rat)	LD50 = 5000 mg/kg (Rabbit)	LC50 = 11700 ppm (Rat) 4 h
Isopropyl alcohol	5045 mg/kg (Rat) 3600 mg/kg (Mouse)	12800 mg/kg (Rat)	72.6 mg/L (Rat) 4 h
Bitrex	LD50 = 584 mg/kg (Rat)	LD50 > 2000 mg/kg (Rat)	LC50 = 0.2 mg/L (Rat) 4 h

(b) skin corrosion/irritation;

Based on available data, the classification criteria are not met

(c) serious eye damage/irritation;

Category 2

(d) respiratory or skin sensitization;

Respiratory

Based on available data, the classification criteria are not met

Skin

Based on available data, the classification criteria are not met

Component	Test method	Test species	Study result
Ethyl alcohol 64-17-5 (>95)	Mouse Ear Swelling Test (MEST)	mouse	non-sensitising

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	OECD Test Guideline 429 Local Lymph Node Assay	mouse	non-sensitising
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(e) germ cell mutagenicity; Based on available data, the classification criteria are not met

Component	Test method	Test species	Study result
Ethyl alcohol 64-17-5 (>95)	AMES test OECD Test Guideline 471	in vitro Bacteria	negative
	Gene cell mutation OECD Test Guideline 476	in vitro Mammalian	negative

(f) carcinogenicity; Based on available data, the classification criteria are not met
The table below indicates whether each agency has listed any ingredient as a carcinogen

(g) reproductive toxicity; Based on available data, the classification criteria are not met

Component	Test method	Test species / Duration	Study result
Ethyl alcohol 64-17-5 (>95)	OECD Test Guideline 416	Oral / mouse 2 Generation	NOAEL = 13.8 g/kg/day
	OECD Test Guideline 414	Inhalation / Rat	NOAEC = 16000 ppm

(h) STOT-single exposure; Based on available data, the classification criteria are not met

Results / Target organs Central nervous system (CNS).

(i) STOT-repeated exposure; Based on available data, the classification criteria are not met

Target Organs None known.

(j) aspiration hazard; Based on available data, the classification criteria are not met

Symptoms / effects, both acute and delayed Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.

11.2. Information on other hazards

Endocrine Disrupting Properties
Assess endocrine disrupting
properties for human health

Contains a substance on the National Authorities Endocrine Disruptor Lists

Component	EU National Authorities Endocrine Disruptor Lists - Health
Methyl ethyl ketone 78-93-3 (1)	List II

SECTION 12: ECOLOGICAL INFORMATION

12.1. Toxicity

Ecotoxicity effects

Contains a substance which is: Toxic to aquatic organisms. The product contains following substances which are hazardous for the environment.

Component	Freshwater Fish	Water Flea	Freshwater Algae
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Ethyl alcohol	Fathead minnow (Pimephales promelas) LC50 = 14200 mg/L/96h	EC50 = 9268 mg/L/48h EC50 = 10800 mg/L/24h	EC50 (72h) = 275 mg/l (Chlorella vulgaris)
Methyl ethyl ketone	Lepomis macrochirus: LC50=3,22 g/L 96 h	EC50: = 5091 mg/L, 48h (Daphnia magna) EC50: 4025 - 6440 mg/L, 48h Static (Daphnia magna) EC50: > 520 mg/L, 48h (Daphnia magna)	
Isopropyl alcohol	LC50: = 9640 mg/L, 96h flow-through (Pimephales promelas) LC50: > 1400000 µg/L, 96h (Lepomis macrochirus) LC50: = 11130 mg/L, 96h static (Pimephales promelas) LC50: = 10000000 µg/L, 96h (Daphnia)	13299 mg/L EC50 = 48 h 9714 mg/L EC50 = 24 h	EC50: > 1000 mg/L, 72h (Desmodesmus subspicatus) EC50: > 1000 mg/L, 96h (Desmodesmus subspicatus)

Component	Microtox	M-Factor
Ethyl alcohol	Photobacterium phosphoreum:EC50 = 34634 mg/L/30 min Photobacterium phosphoreum:EC50 = 35470 mg/L/5 min	
Methyl ethyl ketone	EC50 = 3403 mg/L 30 min EC50 = 3426 mg/L 5 min	
Isopropyl alcohol	= 35390 mg/L EC50 Photobacterium phosphoreum 5 min	

12.2. Persistence and degradability

Persistence

Persistence is unlikely, based on information available.

Component	Degradability
Ethyl alcohol 64-17-5 (>95)	OECD 301E = 94%
Methyl ethyl ketone 78-93-3 (1)	98% (28d)

Degradation in sewage treatment plant

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

12.3. Bioaccumulative potential

Bioaccumulation is unlikely

Component	log Pow	Bioconcentration factor (BCF)
Ethyl alcohol	-0.32	No data available
Methyl ethyl ketone	0.29	No data available
Isopropyl alcohol	0.05	No data available
Bitrex	2.2	No data available

12.4. Mobility in soil

The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces Will likely be mobile in the environment due to its volatility. Disperses rapidly in air

12.5. Results of PBT and vPvB assessment

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB).

12.6. Endocrine disrupting properties

Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

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12.7. Other adverse effects

Persistent Organic Pollutant
Ozone Depletion Potential

This product does not contain any known or suspected substance
This product does not contain any known or suspected substance

SECTION 13: DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Waste from Residues/Unused Products

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

Contaminated Packaging

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

European Waste Catalogue (EWC)

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

Other Information

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations.

Switzerland - Waste Ordinance

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

SECTION 14: TRANSPORT INFORMATION

IMDG/IMO

<u>14.1. UN number</u>	UN1170
<u>14.2. UN proper shipping name</u>	ETHANOL
<u>14.3. Transport hazard class(es)</u>	3
<u>14.4. Packing group</u>	II

ADR

<u>14.1. UN number</u>	UN1170
<u>14.2. UN proper shipping name</u>	ETHANOL
<u>14.3. Transport hazard class(es)</u>	3
<u>14.4. Packing group</u>	II

IATA

<u>14.1. UN number</u>	UN1170
<u>14.2. UN proper shipping name</u>	ETHANOL
<u>14.3. Transport hazard class(es)</u>	3
<u>14.4. Packing group</u>	II

14.5. Environmental hazards No hazards identified

14.6. Special precautions for user No special precautions required.

14.7. Maritime transport in bulk according to IMO instruments Not applicable, packaged goods

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SECTION 15: REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

Component	CAS No	EINECS	ELINCS	NLP	IECSC	TCSI	KECL	ENCS	ISHL
Ethyl alcohol	64-17-5	200-578-6	-	-	X	X	KE-13217	X	X
Methyl ethyl ketone	78-93-3	201-159-0	-	-	X	X	KE-24094	X	X
Isopropyl alcohol	67-63-0	200-661-7	-	-	X	X	KE-29363	X	X
Bitrex	3734-33-6	223-095-2	-	-	X	X	KE-11775	X	-

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	DSL	NDSL	AICS	NZIoC	PICCS
Ethyl alcohol	64-17-5	X	ACTIVE	X	-	X	X	X
Methyl ethyl ketone	78-93-3	X	ACTIVE	X	-	X	X	X
Isopropyl alcohol	67-63-0	X	ACTIVE	X	-	X	X	X
Bitrex	3734-33-6	X	ACTIVE	X	-	X	X	X

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

Authorisation/Restrictions according to EU REACH

Component	CAS No	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Ethyl alcohol	64-17-5	-	-	-
Methyl ethyl ketone	78-93-3	-	Use restricted. See item 75. (see link for restriction details)	-
Isopropyl alcohol	67-63-0	-	Use restricted. See item 75. (see link for restriction details)	-
Bitrex	3734-33-6	-	-	-

REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

Seveso III Directive (2012/18/EC)

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements
Ethyl alcohol	64-17-5	Not applicable	Not applicable
Methyl ethyl ketone	78-93-3	Not applicable	Not applicable
Isopropyl alcohol	67-63-0	Not applicable	Not applicable
Bitrex	3734-33-6	Not applicable	Not applicable

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

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Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

WGK Classification Water endangering class = 1 (self classification)

Component	Germany - Water Classification (AwSV)	Germany - TA-Luft Class
Ethyl alcohol	WGK1	
Methyl ethyl ketone	WGK1	
Isopropyl alcohol	WGK1	
Bitrex	WGK2	

Component	France - INRS (Tables of occupational diseases)
Ethyl alcohol	Tableaux des maladies professionnelles (TMP) - RG 84
Methyl ethyl ketone	Tableaux des maladies professionnelles (TMP) - RG 84
Isopropyl alcohol	Tableaux des maladies professionnelles (TMP) - RG 84

Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

Component	Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81)	Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC)	Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure
Ethyl alcohol 64-17-5 (>95)		Group I	
Methyl ethyl ketone 78-93-3 (1)		Group I	
Isopropyl alcohol 67-63-0 (1)		Group I	
Bitrex 3734-33-6 (0.001)	Prohibited and Restricted Substances		

15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

SECTION 16: OTHER INFORMATION

Full text of H-Statements referred to under sections 2 and 3

H319 - Causes serious eye irritation

EUH066 - Repeated exposure may cause skin dryness or cracking

H225 - Highly flammable liquid and vapor

H302 - Harmful if swallowed

H315 - Causes skin irritation

H335 - May cause respiratory irritation

H336 - May cause drowsiness or dizziness

Legend

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CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

DNEL - Derived No Effect Level

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

NOEC - No Observed Effect Concentration

PBT - Persistent, Bioaccumulative, Toxic

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer
Predicted No Effect Concentration (PNEC)

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

POW - Partition coefficient Octanol:Water

vPvB - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (volatile organic compound)

Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

Physical hazards On basis of test data

Health Hazards Calculation method

Environmental hazards Calculation method

Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

Chemical incident response training.

Prepared By Health, Safety and Environmental Department

Creation Date 29-Jan-2018

Revision Date 24-Mar-2024

Revision Summary New emergency telephone response service provider.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No
1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2,
Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and
Preparations).**

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

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End of Safety Data Sheet